Machine-Dependent Weighted Chairman Assignment: Counterexamples to Two Conjectures and a 7/6 Lower Bound

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Abstract

Liu and Reis conjectured that every fractional assignment with positive machine-dependent weights admits an integral assignment whose discrepancy on every row and every prefix is at most the largest weight. We give an exact rational counterexample with 11 rows and 15 columns. Every fractional column has support two, all weights belong to $\{1/1000, 1/2, 1\}$, and the largest weight is one. A three-column detector gadget forces one decision in each of five blocks. The five forced decisions make an accumulator row attain discrepancy at least $619/600 > 1$. The proof allows integral assignments outside the fractional support, and therefore refutes the unrestricted machine-dependent conjecture itself as well as the stronger unified support-preserving formulation.

A balanced version of the same gadget strengthens the refutation: for every $\varepsilon > 0$ there are finite instances, still with unit maximum weight and fractional support two per column, on which every integral assignment incurs some row-and-prefix discrepancy at least $7/6 - \varepsilon$; a concrete 49×72 instance forces discrepancy above $9/8$. The constant $7/6$ is optimal for this gadget once the detector split is fixed at one half; releasing the split improves the constant to $4 - 2\sqrt{2} - \varepsilon$, with an exact rational witness in which all three violation heights equal $239/204$. We know of no constant upper bound for machine-dependent prefix discrepancy and pose the determination of the optimal constant as an open problem.

1 Introduction

Let $x \in [0,1]^{m\times n}$ be a fractional assignment with

$$\sum_{i=1}^{m} x_{ij} = 1 \qquad (j = 1, \dots, n).$$

An integral assignment $y \in \{0,1\}^{m\times n}$ satisfies the same column sums. Given positive machine-dependent weights d_{ij}, define the row-prefix discrepancy

$$\Delta_i(t) = \sum_{j=1}^{t} d_{ij}(x_{ij} - y_{ij}).$$

The unweighted case is the classical chairman assignment problem: Tijdeman proved the guarantee 1 together with the matching lower bound, Meijer established the optimal constant $1 - 1/(2m-2)$ [7], and Tijdeman later gave an alternative algorithmic proof [9]. Liu and Reis proved the weighted (common column weight) generalization — confirming a special case of the two-sided unsplittable-flow conjecture of Morell and Skutella [8] — and proposed the following machine-dependent extension [6].

Conjecture 1 (Liu–Reis, Conjecture 19)**.** For every fractional assignment x and every positive weight matrix d, there is an integral assignment y such that

$$|\Delta_i(t)| \le D := \max_{i,j} d_{ij} \qquad \text{for every row } i \text{ and prefix } t.$$

Their Conjecture 21 gives a unified statement with support preservation and more general prescribed integral column sums. (Conjecture numbers follow the ITCS 2026 proceedings version [6]; in the arXiv preprint 2511.18546 the same statements appear as Conjectures 3 and 5.) We show that already Conjecture 1 is false.

Theorem 2. *There is a rational instance with $m = 11$, $n = 15$, and $D = 1$ for which every integral assignment y has*

$$|\Delta_i(t)| > 1$$

for some row i and prefix t. Every fractional column has support two and every weight is strictly positive.

Since the theorem excludes all integral assignments, including those outside the support of x, it immediately implies the following.

Corollary 3. *The unified support-preserving statement of Liu and Reis (Conjecture 21) is false, already in the one-winner-per-column case.*

2 The instance

The rows are

$$B, A_1, H_1, A_2, H_2, \ldots, A_5, H_5,$$

and the columns occur in five consecutive blocks

$$J_k, K_k, L_k \qquad (k = 1, \ldots, 5).$$

Set

$$\eta = \frac{1}{1000}.$$

Every fractional entry not listed below is zero, and every weight not listed below equals η. Thus all weights are strictly positive.

For each block k, define the nonzero fractional entries and exceptional weights by Table 1.

Table 1: One block of the counterexample. Here $A = A_k$ and $H = H_k$.

column	nonzero fractional entries	exceptional weights
J_k	$x_{A,J_k} = 19/24$, $x_{B,J_k} = 5/24$	$d_{A,J_k} = d_{B,J_k} = 1$
K_k	$x_{A,K_k} = 1/2$, $x_{H,K_k} = 1/2$	$d_{A,K_k} = 1/2$, $d_{H,K_k} = 1$
L_k	$x_{A,L_k} = 23/48$, $x_{H,L_k} = 25/48$	$d_{A,L_k} = d_{H,L_k} = 1$

Every column sums to one, every column has support two, and

$$d_{ij} \in \left\{ \frac{1}{1000}, \frac{1}{2}, 1 \right\}, \qquad D = 1.$$

3 The forcing gadget

Fix a block k. Before column J_k, the fresh rows A_k and H_k have zero fractional entries in every preceding column. An integral assignment may nevertheless have assigned earlier columns to them. There are at most $3(k-1) \leq 12$ earlier columns, and each such off-support assignment contributes exactly $-\eta$. Hence immediately before J_k,

$$-12\eta \leq \Delta_{A_k} \leq 0, \qquad -12\eta \leq \Delta_{H_k} \leq 0. \tag{1}$$

Lemma 4. *Every integral assignment satisfying $|\Delta_i(t)| \leq 1$ on all rows and prefixes must assign J_k to A_k.*

Proof. Suppose that J_k is not assigned to A_k. Then A_k is unassigned on J_k and gains 19/24. By (1), immediately after J_k,

$$\Delta_{A_k} \geq -12\eta + \frac{19}{24}. \tag{2}$$

The row H_k may itself receive J_k outside its support, which costs only η, so

$$\Delta_{H_k} \geq -13\eta. \tag{3}$$

Consider K_k. If it is not assigned to A_k, then A_k is unassigned and gains

$$d_{A_k,K_k} x_{A_k,K_k} = \frac{1}{2} \cdot \frac{1}{2} = \frac{1}{4}.$$

Together with (2), this gives

$$\Delta_{A_k} \geq -12\eta + \frac{19}{24} + \frac{1}{4} = \frac{25}{24} - \frac{12}{1000} > 1.$$

Thus a surviving assignment must send K_k to A_k.

After that forced choice,

$$\Delta_{A_k} \geq -12\eta + \frac{19}{24} - \frac{1}{4} = -12\eta + \frac{13}{24}, \tag{4}$$

$$\Delta_{H_k} \geq -13\eta + \frac{1}{2}. \tag{5}$$

It remains to assign L_k. If L_k is assigned to H_k, then A_k is unassigned and gains 23/48. By (4),

$$\Delta_{A_k} \geq -12\eta + \frac{13}{24} + \frac{23}{48} = \frac{49}{48} - \frac{12}{1000} > 1.$$

If L_k is not assigned to H_k, then H_k is unassigned and gains 25/48. By (5),

$$\Delta_{H_k} \geq -13\eta + \frac{1}{2} + \frac{25}{48} = \frac{49}{48} - \frac{13}{1000} = \frac{6047}{6000} > 1.$$

Every possible assignment of L_k therefore violates the unit bound. The assumption $J_k \not\to A_k$ was impossible. □

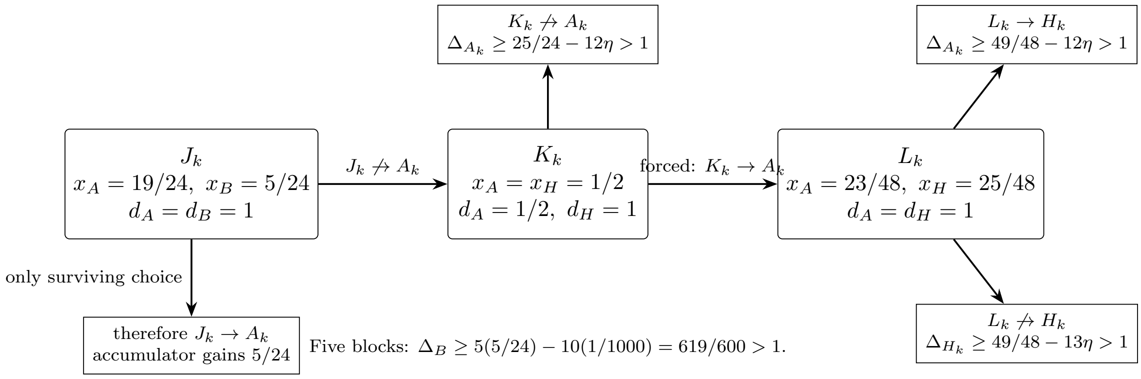

Figure 1: The three-column forcing gadget. The inequalities include the worst possible losses from assignments outside the fractional support.

4 Accumulator contradiction

Apply Lemma 4 independently to all five blocks. Every hypothetical discrepancy-good assignment must satisfy

$$J_k \longmapsto A_k \qquad (k = 1, \dots, 5).$$

The accumulator row B is therefore unassigned on every J_k. Since $x_{B,J_k} = 5/24$ and $d_{B,J_k} = 1$, each block increases its discrepancy by $5/24$.

The only possible decreases occur when one of the ten detector columns K_k, L_k is assigned to B. On each such column $x_{B,j} = 0$ and $d_{B,j} = \eta$, so the decrease is at most η. At the final prefix,

$$\Delta_B(15) \ge 5 \cdot \frac{5}{24} - 10\eta = \frac{25}{24} - \frac{1}{100} = \frac{619}{600} > 1.$$

This contradiction proves Theorem 2.

5 A parametric family

The constants are instances of a simple open parameter region. In one block, choose

	x_A	x_H or x_B	d_A	d_H or d_B
J	$1-p$	p (B)	1	1
K	$1/2$	$1/2$ (H)	a	1
L	$1-q$	q (H)	1	1.

The detector works, before the small η corrections, whenever

$$\frac{a}{2} > p, \qquad \frac{1}{2} < q < 1 - p - \frac{a}{2}.$$

Indeed, the three forcing margins are

$$\frac{a}{2} - p, \qquad 1 - p - \frac{a}{2} - q, \qquad q - \frac{1}{2}.$$

With N blocks the accumulator margin is $Np - 1$, so we also require $Np > 1$. Once these inequalities are strict, $\eta > 0$ can be chosen small enough to absorb every off-support assignment.

Our rational choice is

$$N = 5, \qquad p = \frac{5}{24}, \qquad a = \frac{1}{2}, \qquad q = \frac{25}{48}, \qquad \eta = \frac{1}{1000}.$$

The ideal margins are respectively $1/24, 1/48, 1/48$, while the accumulator margin is $1/24$.

6 A stronger lower bound: the constant $7/6$

The family does more than refute the unit bound. Write

$$m_1 = \frac{a}{2} - p, \qquad m_2 = 1 - p - \frac{a}{2} - q, \qquad m_3 = q - \frac{1}{2}$$

for the three forcing margins, and

$$M(p, a, q) = \min\{m_1, m_2, m_3\}.$$

The three violation heights in the proof of Lemma 4 are then $1 + m_1$, $1 + m_2$, and $1 + m_3$, before the η corrections.

Lemma 5. *Let $c \geq 1$, and suppose an integral assignment satisfies $|\Delta_i(t)| \leq c$ on all rows and prefixes of the N-block instance with parameters (p, a, q, η). If*

$$1 + M(p, a, q) - (3N - 2)\eta > c,$$

then the assignment sends J_k to A_k for every block k. If moreover $Np - 2N\eta > c$, no such assignment exists; that is, every integral assignment has some row and prefix with

$$\Delta_i(t) \;\geq\; \min\bigl\{\, 1 + M(p, a, q) - (3N - 2)\eta,\;\; Np - 2N\eta \,\bigr\}.$$

Proof. The case analysis is that of Lemma 4 verbatim, with the unit bound replaced by c. The pre-block drift bound (1) becomes $\Delta \geq -3(N-1)\eta$ for the fresh rows, and H_k may additionally absorb J_k off support, so every displayed lower bound loses at most $(3N-2)\eta$. The three terminal bounds in the three branches become $(1-p) + \frac{a}{2}$, $(1-p) - \frac{a}{2} + (1-q)$, and $\frac{1}{2} + q$, that is, $1 + m_1$, $1 + m_2$, and $1 + m_3$, each reduced by at most $(3N-2)\eta$; if each exceeds c, all three branches contradict the budget, forcing $J_k \mapsto A_k$. The accumulator argument of Section 4 then gives $\Delta_B(3N) \geq Np - 2N\eta > c$, the final contradiction. The displayed guarantee follows by applying the argument with c equal to the maximum prefix discrepancy of the given assignment. $\square$

Theorem 6. *For every $\varepsilon > 0$ there is a finite instance with strictly positive machine-dependent weights, maximum weight $D = 1$, and every fractional column of support two, on which every integral assignment y satisfies*

$$|\Delta_i(t)| \;\geq\; \frac{7}{6} - \varepsilon$$

for some row i and prefix t.

Proof. Balance the margins. For $p \in (0, \frac{1}{4})$ set

$$a = \frac{1}{3} + \frac{2p}{3}, \qquad q = \frac{2}{3} - \frac{2p}{3}.$$

Then

$$m_1 = m_2 = m_3 = \frac{1}{6} - \frac{2p}{3} \;>\; 0, \qquad 1 + M = \frac{7}{6} - \frac{2p}{3}.$$

Assume without loss of generality $\varepsilon \le \frac{7}{6}$, and choose

$$p = \min\Big\{\frac{3\varepsilon}{8},\ \frac{1}{8}\Big\}, \qquad N = \Big\lceil \frac{2}{p} \Big\rceil, \qquad \eta = \frac{\varepsilon}{13N}.$$

If $\varepsilon < \frac{1}{3}$ then $\frac{2p}{3} = \frac{\varepsilon}{4}$; otherwise $p = \frac{1}{8}$ and $\frac{2p}{3} = \frac{1}{12} \le \frac{\varepsilon}{4}$. In both cases

$$\frac{2p}{3} + (3N-2)\eta \ \le\ \frac{\varepsilon}{4} + 3N \cdot \frac{\varepsilon}{13N} \ =\ \frac{25\varepsilon}{52} \ <\ \varepsilon,$$

so $1 + M - (3N-2)\eta > \frac{7}{6} - \varepsilon$, and

$$Np - 2N\eta \ \ge\ 2 - \frac{2\varepsilon}{13} \ \ge\ 2 - \frac{7}{39} \ >\ \frac{7}{6}.$$

By Lemma 5, every integral assignment has a prefix discrepancy at least $\min\{1 + M - (3N-2)\eta,\ Np - 2N\eta\} > \frac{7}{6} - \varepsilon$. All weights lie in $\{\eta, a, 1\} \subset (0, 1]$, so $D = 1$, and every column retains fractional support two. $\square$

Remark 7. The constant 7/6 is optimal for this three-column family *as parameterized*, that is, with the detector split fixed at $x_{A,K} = x_{H,K} = \frac{1}{2}$. Since $m_2 + m_3 = \frac{1}{2} - p - \frac{a}{2}$,

$$M \le \min\Big\{\frac{a}{2} - p,\ \frac{1}{4} - \frac{p}{2} - \frac{a}{4}\Big\},$$

and the two bounds balance at $a = \frac{1}{3} + \frac{2p}{3}$ with common value $\frac{1}{6} - \frac{2p}{3} < \frac{1}{6}$. Hence $1 + M < \frac{7}{6}$ for every admissible (p, a, q), and the supremum 7/6 is approached only as $p \downarrow 0$, which forces the number of blocks to grow. The half split itself is not optimal; see Remark 8.

Remark 8 (An improved constant from the same three columns)**.** Free the detector split. In the limit $p \downarrow 0$, take block columns J, K, L with supports as before, detector weights $d_{A,K} = d \in (0, 1]$, $d_{H,K} = 1$, and splits $x_{A,K} = x_2$, $x_{A,L} = x_3$. The three violation heights of the forcing argument become

$$v_1 = 1 + dx_2, \qquad v_A = 1 - d(1 - x_2) + x_3, \qquad v_H = (1 - x_2) + (1 - x_3).$$

Since v_1 increases in x_2 while v_H decreases in both splits and v_A increases in x_3, the max–min is attained where all three heights are equal: $x_3 = d$ and $x_2 = \frac{1-d}{1+d}$, with common value

$$\tau(d) = 1 + \frac{d(1-d)}{1+d},$$

maximized at $d = \sqrt{2} - 1$ with value $\sup_d \tau(d) = 4 - 2\sqrt{2} \approx 1.1716$. The rational choice $d = \frac{5}{12}$ gives $x_2 = \frac{7}{17}$, $x_3 = \frac{5}{12}$, and all three heights exactly $\frac{239}{204} > \frac{7}{6}$. For $p > 0$ the heights v_1 and v_A each decrease by p while v_H is unchanged, so Lemma 5 applies with the balanced value $1 + M$ replaced by $\tau(d) - p$; letting $p \downarrow 0$ absorbs the loss exactly as in the proof of Theorem 6. Hence for every $\varepsilon > 0$ some finite instance forces discrepancy at least $4 - 2\sqrt{2} - \varepsilon$. Deeper detector blocks improve the constant further (the discussion records a depth-four witness); a systematic treatment is deferred to follow-up work.

Corollary 9. *The concrete choice*

$$N = 24, \qquad p = \frac{1}{20}, \qquad a = \frac{11}{30}, \qquad q = \frac{19}{30}, \qquad \eta = \frac{1}{10000}$$

gives an instance with $m = 49$ rows, $n = 72$ columns, and weights in $\{10^{-4},\ 11/30,\ 1\}$ on which every integral assignment has some row-and-prefix discrepancy at least

$$\frac{3379}{3000} > \frac{9}{8}.$$

Proof. Here $m_1 = m_2 = m_3 = \frac{2}{15}$, so $1 + M = \frac{17}{15}$, and

$$1 + M - (3N-2)\eta = \frac{17}{15} - \frac{70}{10000} = \frac{3379}{3000}, \qquad Np - 2N\eta = \frac{6}{5} - \frac{48}{10000} = \frac{11952}{10000},$$

whose minimum is $\frac{3379}{3000} > \frac{3375}{3000} = \frac{9}{8}$. Apply Lemma 5. $\square$

7 Discussion

The counterexample exploits exactly what changes when a common job weight is replaced by machine-dependent weights. With a common weight, assigning a job redistributes a conserved amount among rows. Here the K column has weight $1/2$ on A_k but weight 1 on H_k, permitting the detector to amplify a prefix imbalance. The small value η keeps every off-support weight positive without allowing unsupported choices to repair the forcing chain.

The result does not address the ordinary weighted-carpooling conjecture, in which a column has one common weight across all rows. It rules out the machine-dependent extension and every stronger statement containing it.

An open problem. Theorem 6 makes the failure quantitative, and Remarks 7 and 8 locate the exact reach of the three-column gadget. Let κ denote the supremum, over all finite instances with positive machine-dependent weights, of the minimum over integral assignments of the maximum row-and-prefix discrepancy divided by $D = \max_{i,j} d_{ij}$. We have shown $\kappa \geq 4 - 2\sqrt{2}$, and deeper detector blocks push the bound higher still: a depth-four detector with weights $(9/25, 3/5)$ admits an exact rational certificate with all forcing heights equal to $103/85 > 4 - 2\sqrt{2}$, to appear in follow-up work. In the opposite direction we are aware of no finite upper bound on κ at all; per instance nothing improves on the trivial bound of $n \cdot D$ on the discrepancy itself, which gives no bound on the supremum κ. The non-prefix version of the machine-dependent statement does hold [6], by the rounding argument of Lenstra, Shmoys, and Tardos [5], and for the common-weight prefix problem the best known bounds are $2m \cdot \max_j d_j$, via the linear-algebraic argument of Bárány and Grinberg [3, 4], and the $O(\sqrt{\log n}) \cdot \max_j d_j$ bound of Bansal, Rohwedder, and Svensson [2] via Banaszczyk's theorem on series of signed vectors [1]; whether either argument survives machine-dependent weights appears to be open. Determining whether κ is finite, and if so its value, is the natural next question.

Reproducibility and disclosure

The accompanying repository contains the full 11×15 rational instance as JSON, an exact `Fraction`-based verifier for every inequality in the forcing certificate, and an independent mixed-integer infeasibility check. For Section 6 it further contains an exact verifier for the balanced-family certificate of Lemma 5 (including the 49×72 instance of Corollary 9 and sampled values of ε in Theorem 6), and an independent mixed-integer check that the 49×72 instance admits no assignment with all prefix discrepancies at most $9/8$. A generative AI system (GPT-5.6 Pro) assisted with exploration, checking, code drafting, and exposition. It is not an author. The author assumes responsibility for independent verification, priority, attribution, and the final submitted text.

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